Léopold Maillard

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Education

École Polytechnique — Palaiseau, France
PhD in Mathematics and Computer Science — *2024 – Present*
- Advised by Prof. Maks Ovsjanikov in the GeomeriX group at the LIX laboratory.
- Thesis: *Generative Models for Guided Synthesis of Complex and Functional 3D Scenes*.
- Expected defense: January 2027.

INSA Rouen Normandie — Saint-Étienne-du-Rouvray, France
Diplôme d'Ingénieur – Bachelor & Master of Engineering in Computer Science — *2017 – 2022*

Université de Rouen Normandie — Saint-Étienne-du-Rouvray, France
Master of Science in Data Science | *Ranked first* — *2021 – 2022*

Experience

Research Scientist in 3D Machine Learning — Sep. 2022 – Present
Dassault Systèmes | *Corporate Research* — *Paris, France*
- Since 2024, conducting an industrial PhD with École Polytechnique within this role, leading to first-author conference papers at NeurIPS [C.1] and ICCV [C.2].
- Developed DeBaRA [C.1], a state-of-the-art diffusion-based 3D layout generation model; packaged and transferred its codebase to the R&D team of a leading interior design and room planning platform for product integration.
- Developed LACONIC [C.2], a lightweight adapter that conditions pretrained text-to-image diffusion models on 3D scene layouts, enabling explicit camera control and object-level editing over spatial and semantic features.
- Currently developing a unified flow-matching framework jointly modeling 2D images and 3D scene structure.
- Designed and prototyped 6 patent-pending methods related to 3D layout synthesis and controllable image generation.
- Mentored 4 research interns on projects that resulted in publications and patent applications.

Visiting Student Researcher — Jun. 2025 – Sep. 2025
Stanford University | *Computer Science Department* — *Stanford, CA, USA*
- Advised by Prof. Leonidas Guibas and Prof. Francis Engelmann.
- Developed SceneTeract [S.1], a tool-augmented verification interface checking whether embodied agents can carry out activities in 3D indoor scenes. Post-trained a VLM with GRPO on its feedback to improve feasibility assessment.

Research Intern in CAD Machine Learning — Mar. 2022 – Sep. 2022
Dassault Systèmes | *Corporate Research* — *Paris, France*
- Designed a patent-pending deep architecture to automate the assembly of 3D parts in CAD workflows, implemented and trained from scratch in PyTorch Geometric.

Research Intern — Jun. 2021 – Aug. 2021
New York University | *NYU Courant* — *New York, NY, USA*
- Advised by Prof. Dennis Shasha.
- Built an on-device system estimating meal nutrition via food segmentation and stereo-based volume estimation.

Student Data Scientist — Jan. 2021 – Jan. 2022
Essilor International — *Rouen, France*
- Designed and implemented an end-to-end deep learning pipeline for real-time facial and eyewear landmark detection.

Publications & Preprints

C=Conference, J=Journal, S=In Submission, T=Thesis

[C.1] **L. Maillard**, N. Sereyjol-Garros, T. Durand, and M. Ovsjanikov, "DeBaRA: Denoising-Based 3D Room Arrangement Generation," in *Advances in Neural Information Processing Systems (**NeurIPS**)*, 2024.

[C.2] **L. Maillard**, T. Durand, A. Ramanana Rahary, and M. Ovsjanikov, "LACONIC: A 3D Layout Adapter for Controllable Image Creation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (**ICCV**)*, 2025.

[C.3] V.Y. Chen, E. Pierson, **L. Maillard**, and M. Ovsjanikov, "Beyond Prompts: Unconditional 3D Inversion for Out-of-Distribution Shapes," in *European Conference on Computer Vision (**ECCV**)*, 2026.

[S.1] **L. Maillard**, F. Engelmann, T. Durand, B. Pan, Y. You, L.J. Guibas, and M. Ovsjanikov, "SceneTeract: Probing and Improving Agent-Aware Activity Reasoning in 3D Indoor Scenes," *under review*, 2026.

Selected Patent Applications

F=Filed, P=Published, G=Granted

[F.1] **L. Maillard** and T. Durand, "Generating 2D Image of 3D Scene with Pre-Trained Conditional Generative Image Model," *European Patent Application EP25306031.3*, 2025.

[P.1] **L. Maillard**, T. Durand, and M. Ovsjanikov, "Conditioned 3D Layout Prediction," *U.S. Patent Application 19/337,687*, 2024.

[P.2] A. Ramanana Rahary, **L. Maillard**, and T. Durand, "Generating 2D Image of 3D Scene with Conditioning Signal," *U.S. Patent Application 19/259,930*, 2024.

[P.3] **L. Maillard** and J. Boucher, "Machine Learning for CAD Model Retrieval," *U.S. Patent Application 18/611,942*, 2023.

Academic Activities

Invited Talks

- **Technion – Israel Institute of Technology**. *Creating Functional Spaces: Generation, Control, and Interactions in 3D Indoor Scenes*. Host: Prof. Or Litany, Mar. 2026.
- **Stanford Geometric Computation Group**. *An Evaluation Framework for Embodied Interactions in 3D Scenes*. Host: Prof. Leonidas Guibas, Sep. 2025.

Lab Teaching Assistant

- **Master MVA, ENS Paris-Saclay**. *Geometry Processing and Geometric Deep Learning* [MVA-GEOM]. *Fall 2026*
- **Université du Québec à Montréal**. *Object-Oriented Programming* [INF2120]. *Winter 2020*

Conference Reviewer

- IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026. *Outstanding Reviewer (Top 5%)*
- Conference on Neural Information Processing Systems (**NeurIPS**), 2026.
- International Conference on 3D Vision (**3DV**), 2027.

Technical Skills

Expertise: Generative Models (Diffusion, Flow Matching, VAEs), Controllable Generation (Conditioning, Guidance, Adapter Networks), Image Generation Backbones (MMDiT, UNet), Parameter-Efficient Fine-Tuning (LoRA), Vision-Language Models (VLMs), RL Post-Training (GRPO), 3D Deep Learning, 3D Scene Understanding.
Frameworks: PyTorch (incl. multi-GPU training), Hugging Face (Transformers, Diffusers, TRL), Lightning, Hydra.
Programming & Tools: Python, Git, Docker, Linux, TensorBoard, LaTeX.
Languages: French (Native), English (Professional, TOEIC: 955/990), German (Elementary).